Introduction to Robotics, Homework 1

The AR-601M arm

Forward and inverse kinematics, Jacobian, singularities

MIPT

Center for Cognitive Modeling

What this homework is

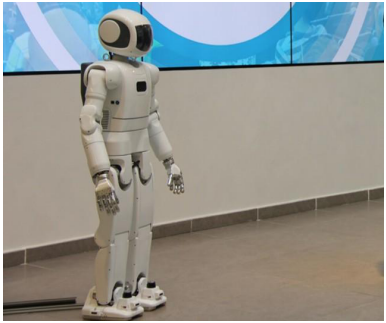
One arm, six tasks, eighteen hours

AR-601M is a 41-DOF humanoid robot. This homework treats one of its two arms in isolation: a 6-DOF serial chain, shoulder to tool, with a spherical wrist.

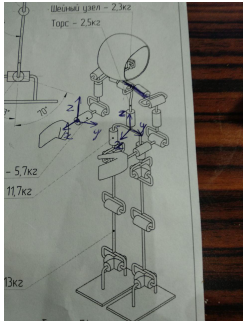
- **Forward** kinematics: joint angles → tool pose.
- **Inverse** kinematics, analytical and numerical: tool pose → joint angles.
- The **Jacobian**, and where it loses rank: **singularities**.

About 18 hours of work covering weeks 1–3. This deck is a pointer; the full statement, with every derivation, requirement and hint, is the handout linked on the last frame.

The mechanism: AR-601M



(a) The AR-601M arm.

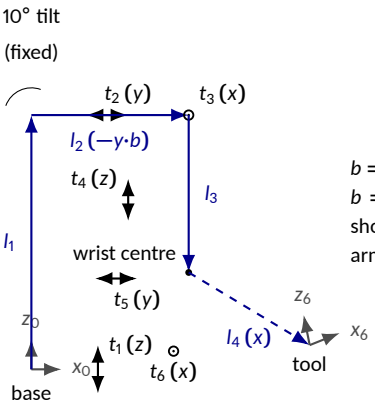


(b) The blueprint, annotated in blue.

Figure: The robot, and the blueprint it is modelled from; lengths are not legible on the blueprint, see handout §2.

The kinematic scheme and the chain

$$^{0}T_6 = R_z(t_1)\,T_z(l_1)\,R_z(10°b)\,R_y(t_2)\,T_y(-l_2 b)\,R_x(t_3)\,T_z(-l_3)\,R_z(t_4)\,R_y(t_5)\,R_x(t_6)\,T_x(l_4)$$



$b = +1$ shown (right arm).
$b = -1$ mirrors l_2 and the shoulder tilt about z (left arm).

What you will be able to do

- Derive and implement forward kinematics for a serial chain with a spherical wrist.
- Solve an inverse-kinematics target analytically, in closed form, keeping every branch, not just the first one found.
- Solve the same target numerically and reconcile the answer against the analytical branches.
- Build a manipulator Jacobian two independent ways and locate where it loses rank.

The six tasks

- **1.** Kinematic model
- **2.** FK: angles → arm and pose
- **3.** IK: pose → every valid arm
- **4.** Numerical inverse kinematics
- **5.** Jacobian
- **6.** Singularity and branch-count maps

Visualize Task 2's arm and transform, Task 3's unique valid branches, and Task 6's conditioning and fixed-orientation branch counts. Full requirements are in handout §4.

What you hand in

The packaging contract

- A `pixi.toml` or a `Dockerfile` at the repository root.
- One command reproduces the report's numbers, example images and maps from a clean checkout.
- Three functions, exactly these roles: `FK(q, robot)`, `IK(T, robot)`, `Jacobian(q, robot)` — any language, the same three signatures.

Packaging is what is graded, not the language used to write it.

The full statement

Everything above is a pointer. The handout has every derivation, every requirement and every hint.

Read handout §3 and §4 before writing any code.

hw01_handout.pdf on GitHub



The handout

Deserve “A” grade!

– Oleg Bulichev

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Цифра, 521